

## **CO-2 AT-1**

### **Assignment Title: Scenario-Based Requirement Analysis**

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### **Blockchain-Based Academic Credential Verification Platform**

#### **1. Study and Analyze the Scenario**

##### **1.1 Introduction**

Academic certificates are essential documents used for employment, higher education, scholarships, and professional certifications. Traditional verification methods are manual, time-consuming, and vulnerable to document forgery. The proposed Blockchain-Based Academic Credential Verification Platform provides a secure digital system where certificates are issued, stored, and verified using blockchain technology. Each certificate is digitally signed and its hash is stored on the blockchain, ensuring authenticity and preventing tampering.

##### **1.2 Business Domain**

The proposed system belongs to the Education Technology (EdTech) domain and is applicable to:

- Universities and Colleges
- Government Education Departments
- Recruitment Organizations
- Scholarship Verification
- Professional Certification Agencies

##### **1.3 Existing Challenges**

The current system faces several challenges:

- Manual certificate verification
- Fake academic certificates

- High administrative workload
- Slow verification process
- Risk of certificate tampering

## 1.4 System Objectives

The proposed platform aims to:

- Issue secure digital certificates.
- Prevent certificate forgery.
- Enable instant verification.
- Maintain tamper-proof records.
- Support multiple institutions.
- Generate verification reports.

## 2. Functional and Non-Functional Requirements

### 2.1 Functional Requirements

- User Registration and Login
- Student Management
- Certificate Issuance
- Blockchain Storage
- QR Code Generation
- Certificate Verification
- Report Generation
- Audit Trail

### 2.2 Non-Functional Requirements

- **Security:** Encryption, blockchain, digital signatures.
- **Performance:** Fast certificate verification.
- **Reliability:** High availability.
- **Scalability:** Support multiple institutions.
- **Usability:** User-friendly interface.

### **3. Identify Actors and Use Cases**

#### **3.1 Primary Actors**

- Student
- University Administrator
- Employer
- System Administrator

#### **3.2 Secondary Actors**

- Government Authority
- Blockchain Network
- Database Server
- Email Notification Service

#### **3.3 Major Use Cases**

##### **Student**

- Register
- Login
- View and Download Certificate
- Share Certificate

##### **University**

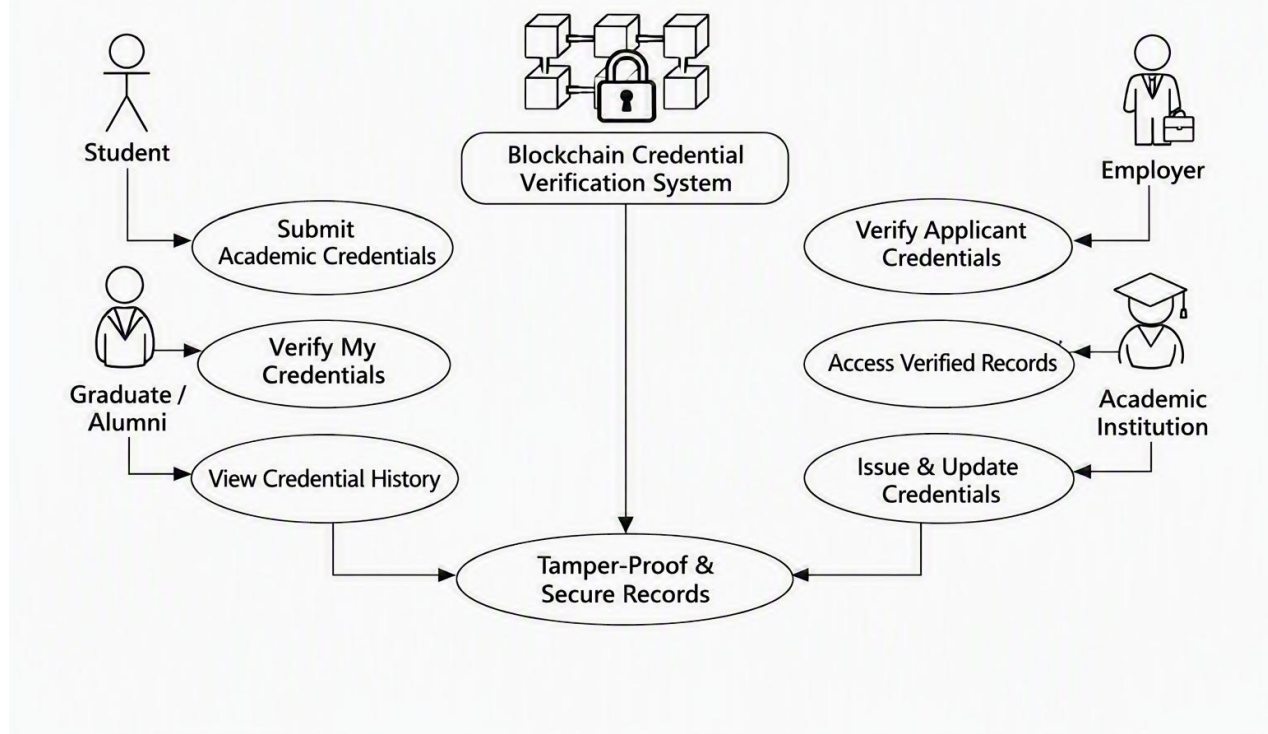
- Register Student
- Issue Certificate
- Update Certificate
- View Reports

##### **Employer**

- Verify Certificate
- Scan QR Code
- Download Verification Report

##### **Administrator**

- Manage Users
- Manage Institutions
- Generate Reports
- Monitor Audit Logs



## 4. Software Requirement Specification (SRS)

### 4.1 Introduction

The proposed platform improves the security and efficiency of issuing and verifying academic certificates. It replaces manual verification with a blockchain-based system that enables secure and instant verification.

### 4.2 Problem Description

Traditional certificate verification is slow, manual, and vulnerable to forgery. The proposed platform addresses these issues by storing certificate information securely on the blockchain.

### 4.3 Scope of the System

The platform includes:

- Certificate Issuance
- Blockchain Storage
- Student Portal
- Employer Verification
- QR Code Verification

- Report Generation
- Audit Management

#### **4.4 Functional Requirements**

The system shall:

- Register and authenticate users.
- Issue digital certificates.
- Store certificate hashes on blockchain.
- Generate QR codes.
- Verify certificates.
- Generate reports.
- Maintain audit logs.

#### **4.5 Non-Functional Requirements**

The system shall provide:

- High Security
- Good Performance
- Reliability
- Scalability
- Maintainability
- User-Friendly Interface

#### **4.6 Use Case Descriptions**

**Use Case 1 – Issue Certificate**

**Primary Actor:** University Administrator

**Precondition:** Student details are verified.

**Flow**

1. Login
2. Select Student
3. Generate Certificate
4. Store Hash on Blockchain
5. Generate QR Code

**Postcondition**

Student receives a verified digital certificate.

## **Use Case 2 – Verify Certificate**

**Primary Actor:** Employer

**Precondition:** Employer has the Certificate ID or QR Code.

### **Flow**

1. Open Verification Portal
2. Enter Certificate ID
3. Verify using Blockchain
4. Display Result

### **Postcondition**

Certificate authenticity is confirmed.

## **4.7 Data Requirements**

### **Student Data**

- Student ID
- Name
- Degree
- Department

### **Certificate Data**

- Certificate ID
- Certificate Hash
- QR Code

### **Institution Data**

- University Name
- Institution ID

### **Verification Data**

- Verification Date
- Status
- Transaction ID

## **4.8 External Interface Requirements**

### **User Interface**

- Web Portal
- Mobile Support

### **Hardware Interface**

- Computer
- Smartphone

### **Software Interface**

- Blockchain Network
- Database

### **Communication Interface**

- HTTPS
- REST APIs

## **4.9 System Features**

- Secure Login
- Certificate Issuance
- Blockchain Verification
- QR Code Verification
- Report Generation
- Audit Trail

## **5. Assumptions and Constraints**

### **5.1 Assumptions**

- Universities maintain accurate records.
- Internet connectivity is available.
- Blockchain network is operational.
- Users follow security guidelines.

### **5.2 Constraints**

#### **Technical**

- Blockchain integration complexity.

#### **Operational**

- User training required.

#### **Economic**

- Initial implementation cost.

#### **Legal**

- Data privacy regulations.

#### **Time**

- Fixed project schedule.

## **6. Validate Requirement Completeness and Consistency**

### **Requirement Validation**

- Completeness: All major stakeholder requirements are included.
- Consistency: No conflicting requirements exist.
- Feasibility: Requirements are technically achievable.
- Traceability: Requirements align with stakeholder needs.

### **Suggested Improvements**

- AI-based fraud detection
- Mobile application
- Biometric authentication
- Government database integration

## **Conclusion**

The Blockchain-Based Academic Credential Verification Platform provides a secure, scalable, and transparent solution for issuing and verifying academic credentials. By leveraging blockchain technology, digital signatures, and QR-code-based verification, the system eliminates certificate fraud, reduces verification time, and improves trust among universities, students, employers, and government authorities. The proposed SRS defines the functional and non-functional requirements, identifies all stakeholders and use cases, specifies system interfaces, and validates requirement completeness, making it a strong foundation for the subsequent design and implementation phases.